

PHASE 2 – RESEARCH & MODEL SELECTION

Project: LegalEase – AI-Powered Legal Document Generator

1. Research Requirements

The project requires knowledge of FastAPI, Google Gemini API, Streamlit, Python, and the required supporting libraries.

2. AI Model Research

The project documentation describes evaluating Gemini models for structured legal document generation. Gemini 1.5 Pro was selected in the source documentation for its long context capability, structured prompting, and ability to generate formal structured text.

3. Selected Model

Selected model: Gemini 1.5 Pro.

4. Technology Selection

Frontend: Streamlit

Backend: FastAPI

AI Core: Gemini integration

Document generation: python-docx and FPDF

Supporting tools: Pillow, requests, python-dotenv

5. System Architecture

The architecture consists of a Streamlit frontend, FastAPI backend, Gemini AI core, and document-formatting modules. The frontend collects inputs, the backend routes requests, the AI core generates content, and formatting modules create TXT, DOCX, and PDF outputs.

6. Development Environment

Python 3.10+ with pip and a virtual environment is specified. The project also requires installation of FastAPI, Uvicorn, Streamlit, Google Generative AI SDK, python-docx, FPDF, Pillow, requests, and python-dotenv.

Source basis: LegalEase project documentation supplied for this task.